ADVANCED PROBLEMS FOR LECTURE 21

In the lecture, we will be talking at a more general level of the patterns of what is going on. But there are deeper questions to explore therein, which we cannot get to. This is your opportunity to investigate deeper for yourself. Don't limit yourself to just these questions, let your curiosity guide you too! That is, if you wonder about any details, or if any certain patterns or ideas exist, see if you can state your question clearly, and answer it! Here are a few suggestions to get you started.

(1) We could certainly use the method of disks or washers to evaluate solids of revolution about the y-axis, just integrate with respect to y instead! But we see that we hit a problem in that we can't easily calculate radii. Is there any nice property of a function that would allow for the method of disks or washers to be used instead of cylindrical shells?

(2) Suppose we want to generalize our revolution to revolving about an axis $y = mx + b$. Can you come up with a general approach to measure the volume you get by rotating the area about this general axis?

(3) Our method of stacking slices to calculate the volume of a three dimensional object amounted to knowing the cross-sectional areas at each x value, and then integrating to "stack" or "add" them up. But we also notice that we can calculate areas by doing another integral. This implies that we can perform an integration within an integration (also known as a double integral) to get a volume.

(a) Write the volume of a box as a double integral. (Find the cross sectional area for a given x-value as an integral, and then integrate over x.)

(b) We will try to do the same thing for cylinders and spheres, but we need to know how to calculate the area of a circle first. Note that the equation of a circle can be given by $x^2 + y^2 = r^2$, so set up an integral over x between the curves $y = \sqrt{r^2 - x^2}$ and $y = -\sqrt{r^2 - x^2}$.

(c) If you attempted to integrate, you may have found it challenging. This may be because the techniques we have developed so far do not suffice. This integral is easier to compute via *trigonometric substitution*. Notice that it must be the case that $x^2 \leq r^2$, so $\frac{x^2}{r^2} \leq 1$. This allows us to substitute $\sin(\theta) = \frac{x}{r}$. Show that our integral becomes

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 2\sqrt{r^2 - r^2\sin^2(\theta)} \cdot r\cos(\theta)d\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} r^2\cos^2(\theta)d\theta$$

(d) Prove that the area of a circle of radius $r > 0$ is πr^2. Use this to calculate the volume of a cylinder and a sphere as a double-integral. (Hint: you may need a trig identity to get rid of the square on the cosine.)